

Structure of Atom

I. Early Ideas and the Atomic Theory of Matter

- The existence of atoms was proposed by early Indian and Greek philosophers around 400 B.C., who viewed them as fundamental, indivisible building blocks of matter (from Greek 'a-tomio' meaning 'uncut-able' or 'non-divisible'). These early ideas were mere speculations, lacking experimental verification.
- John Dalton, a British school teacher, first proposed the atomic theory of matter on a firm scientific basis in 1808.
 - o Dalton's atomic theory considered the atom as the ultimate particle of matter.
 - o It successfully explained the law of conservation of mass, law of constant composition, and law of multiple proportions.
 - o However, it failed to explain phenomena like substances getting electrically charged when rubbed (e.g., glass or ebonite with silk or fur).

II. Discovery of Sub-atomic Particles

- Towards the end of the nineteenth and beginning of the twentieth century, experiments established that atoms are made of sub-atomic particles: electrons, protons, and neutrons, a concept different from Dalton's view.
- A basic rule for charged particles is that like charges repel and unlike charges attract.

A. Discovery of Electron (Cathode Rays)

- Michael Faraday showed in 1830 that electricity passed through an electrolyte solution caused chemical reactions at electrodes, suggesting the particulate nature of electricity.
- In the mid-1850s, scientists, including Faraday, studied electrical discharge in partially evacuated tubes (cathode ray discharge tubes).
 - o When high voltage was applied, a current flowed from the negative electrode (cathode) to the positive electrode (anode), forming cathode rays.

o These rays, though not visible themselves, could be observed by their effect on phosphorescent materials like zinc sulphide, which glowed when struck by the rays.

o Characteristics of Cathode Rays/Electrons:

- Start from the cathode and move towards the anode.

- Travel in straight lines in the absence of electrical or magnetic fields.

- Behave like negatively charged particles in electrical or magnetic fields, leading to the conclusion that cathode rays consist of negatively charged particles called electrons.

- Their characteristics (electrons) are independent of the electrode material or the nature of the gas in the tube.

o Conclusion: Electrons are basic constituents of all atoms.

B. Charge to Mass Ratio of Electron (e/m_e)

• In 1897, British physicist J.J. Thomson measured the ratio of electrical charge (e) to the mass of electron (m_e) using a cathode ray tube with perpendicular electrical and magnetic fields.

o The amount of deflection of particles depends on:

- Magnitude of the negative charge: Greater charge leads to greater deflection.

- Mass of the particle: Lighter particles deflect more.

- Strength of the electrical or magnetic field: Stronger fields increase deflection.

o Thomson determined the e/m_e value as $1.758820 \times 10^{11} \text{ C kg}^{-1}$.

o Since electrons are negatively charged, the charge on an electron is $-e$.

C. Charge on the Electron

• R.A. Millikan (1906-14) devised the oil drop experiment to determine the charge on electrons.

o He found the charge on the electron to be $-1.6 \times 10^{-19} \text{ C}$.

- o The presently accepted value is $-1.602176 \times 10^{-19} \text{ C}$.
- o The mass of the electron (m_e) was then determined by combining this charge value with Thomson's e/m_e ratio, yielding $9.1094 \times 10^{-31} \text{ kg}$.
- o Millikan concluded that the magnitude of electrical charge (q) on the droplets is always an integral multiple of the fundamental electrical charge (e), i.e., $q = n e$, where $n = 1, 2, 3, \dots$

D. Discovery of Protons and Neutrons

• Protons:

- o Electrical discharge experiments in modified cathode ray tubes led to the discovery of canal rays carrying positively charged particles.
- o Characteristics of positively charged particles:
 - Mass depends on the nature of the gas in the tube (they are positively charged gaseous ions).
 - Charge-to-mass ratio depends on the gas from which they originate.
 - Some carry a multiple of the fundamental unit of electrical charge.
 - Their behavior in magnetic/electrical fields is opposite to that of electrons.
- o The smallest and lightest positive ion was obtained from hydrogen and was named proton in 1919.

• Neutrons:

- o A need for an electrically neutral particle in the atom was recognized.
- o Chadwick discovered neutrons in 1932 by bombarding a thin beryllium sheet with α -particles.
- o Electrically neutral particles with a mass slightly greater than protons were emitted, which he named neutrons.
- Properties of Fundamental Particles:

- o Electron (e): Absolute charge $-1.602176 \times 10^{-19}$ C, relative charge -1 , mass 9.109382×10^{-31} kg (approx. 0 u).
- o Proton (p): Absolute charge $+1.602176 \times 10^{-19}$ C, relative charge $+1$, mass $1.6726216 \times 10^{-27}$ kg (approx. 1 u).
- o Neutron (n): Absolute charge 0 C, relative charge 0, mass 1.674927×10^{-27} kg (approx. 1 u).

III. Atomic Models

- After the discovery of sub-atomic particles, scientists faced major problems like accounting for atomic stability, comparing elemental behavior, explaining molecule formation, and understanding atomic spectra.

A. Thomson Model of Atom

- Proposed by J.J. Thomson in 1898.
- Described the atom as a sphere (radius $\approx 10^{-10}$ m) of uniformly distributed positive charge with electrons embedded within it, like plums in a pudding or seeds in a watermelon.
- Assumed the mass of the atom was uniformly distributed.
- It could explain the overall neutrality of the atom.
- Limitation: It was inconsistent with later experimental results. Thomson received the Nobel Prize in Physics in 1906 for his work on electricity conduction in gases.

B. Rutherford's Nuclear Model of Atom

- Ernest Rutherford and his students (Hans Geiger and Ernest Marsden) conducted the α -particle scattering experiment.
 - o They directed a stream of high-energy α -particles (helium nuclei, carrying two units of positive charge and four units of atomic mass) from a radioactive source at a thin gold foil (thickness ≈ 100 nm).
 - o Observations were recorded using a circular fluorescent zinc sulphide screen.
- Expected Results (based on Thomson's model): Most α -particles should pass directly through with small deflections, as

mass was thought to be evenly spread.

• Actual Observations:

- o Most α -particles passed through the gold foil undeflected.
- o A small fraction was deflected by small angles.
- o A very few α -particles (≈ 1 in 20,000) bounced back (deflected by nearly 180°).

• Conclusions:

- o Most of the space in the atom is empty (due to undeflected particles).
- o The positive charge is concentrated in a very small volume at the center, repelling and deflecting positive α -particles (not spread throughout as Thomson presumed). This concentrated region was called the nucleus.
- o Calculations showed the volume occupied by the nucleus is negligibly small compared to the total volume of the atom (atom radius $\approx 10^{-10}$ m, nucleus radius $\approx 10^{-15}$ m; e.g., a cricket ball nucleus in a 5 km atom).

• Rutherford's Nuclear Model Postulates:

- o The positive charge and most mass are densely concentrated in an extremely small region called the nucleus.
- o Electrons move around the nucleus at very high speed in circular paths called orbits (resembling a solar system with nucleus as sun and electrons as planets).
- o Electrons and the nucleus are held together by electrostatic forces of attraction.

C. Atomic Number and Mass Number

- Atomic Number (Z): The number of protons present in the nucleus. It also equals the number of electrons in a neutral atom. (e.g., Hydrogen $Z=1$, Sodium $Z=11$).
- Mass Number (A): The total number of nucleons (protons + neutrons) in the nucleus.
- o Mass Number (A) = Number of protons (Z) + Number of

neutrons (n).

- Representation: An atom can be represented as ${}^Z_A\text{X}$.

- Isobars: Atoms with the same mass number (A) but different atomic number (Z) (e.g., ${}^{14}\text{C}$ and ${}^{14}\text{N}$).

- Isotopes: Atoms with identical atomic number (Z) but different atomic mass number (A).

- o The difference between isotopes is due to different numbers of neutrons.

- o Example: Hydrogen has protium (${}^1\text{H}$, 1 proton, 0 neutron), deuterium (${}^2\text{D}$, 1 proton, 1 neutron), and tritium (${}^3\text{T}$, 1 proton, 2 neutrons).

- o Chemical properties of atoms are controlled by the number of electrons (determined by protons). The number of neutrons has very little effect on chemical properties. Thus, isotopes of a given element show the same chemical behavior.

D. Drawbacks of Rutherford Model

- Instability of Atom: According to Maxwell's electromagnetic theory, an accelerating charged particle (like an electron orbiting the nucleus) should continuously emit electromagnetic radiation.

- o This emission would cause the electron to lose energy and spiral into the nucleus.

- o Calculations suggest an electron should spiral into the nucleus in about 10^{-8} seconds, but atoms are stable.

- o If electrons were stationary, electrostatic attraction would pull them into the nucleus.

- o Therefore, the Rutherford model cannot explain the stability of an atom.

- Electron Distribution: It said nothing about the distribution of electrons around the nucleus or their energies.

IV. Developments Leading to Bohr's Model of Atom

- Niels Bohr's model improved upon Rutherford's by utilizing

two key developments:

- o Dual character of electromagnetic radiation (wave and particle properties).

- o Experimental results regarding atomic spectra.

A. Wave Nature of Electromagnetic Radiation

- James Clerk Maxwell (1870s) first comprehensively explained that accelerating charged particles produce and transmit alternating electrical and magnetic fields in the form of electromagnetic waves/radiation.

- o Light, known from ancient times, was initially thought to be particles (Newton), but its wave nature was established in the 19th century.

- o Properties of Electromagnetic Waves:

- ☐ Oscillating electric and magnetic fields are perpendicular to each other and to the direction of wave propagation.

- ☐ Do not require a medium and can move in vacuum.

- ☐ Vary in wavelength (λ) and frequency (ν), forming the electromagnetic spectrum (e.g., radio waves, microwaves, infrared, visible light, ultraviolet, X-rays).

- ☐ In vacuum, all electromagnetic radiations travel at the same speed, 'c' (speed of light), which is $3.0 \times 10^8 \text{ m s}^{-1}$ (or $2.997925 \times 10^8 \text{ m s}^{-1}$).

- ☐ The relationship: $c = \nu\lambda$.

- ☐ Frequency (ν): Number of waves passing a point per second (SI unit: Hertz, Hz or s^{-1}).

- ☐ Wavelength (λ): Distance between two consecutive crests or troughs (SI unit: meter, m).

- ☐ Wavenumber ($\bar{\nu}$): Number of wavelengths per unit length (units: m^{-1} or cm^{-1}).

B. Particle Nature of Electromagnetic Radiation: Planck's Quantum Theory

- Some phenomena like diffraction (bending of waves) and

interference (combination of waves) could be explained by the wave nature of light.

- However, classical physics could not explain:

- o Black-body radiation (nature of emission from hot bodies).

- o Photoelectric effect (ejection of electrons from metals by light).

- o Variation of heat capacity of solids with temperature.

- o Line spectra of atoms.

- These phenomena suggested that systems take energy only in discrete amounts (quantized).

- Max Planck (1900) provided the first explanation for black-body radiation.

- o He assumed that atoms and molecules emit or absorb energy only in discrete quantities, not continuously.

- o He named the smallest quantity of energy a quantum.

- o The energy (E) of a quantum is proportional to its frequency (ν): $E = h\nu$.

- o ' h ' is Planck's constant, with a value of $6.626 \times 10^{-34} \text{ J s}$.

- o This theory explained the distribution of intensity in black-body radiation.

- o Quantization is like standing on a staircase – you can only be on steps, not in between. Energy can take values of 0, $h\nu$, $2h\nu$, ... $nh\nu$.

C. Photoelectric Effect

- H. Hertz (1887) observed that electrons were ejected when certain metals (e.g., potassium, rubidium, caesium) were exposed to light.

- Observations:

- o Electrons ejected immediately (no time lag).

- o The number of ejected electrons is proportional to the intensity/brightness of light.

- o For each metal, there's a characteristic minimum frequency

(ν_0), called threshold frequency, below which no photoelectric effect occurs.

- o Above ν_0 , ejected electrons have kinetic energy, which increases with increasing frequency of light.

- Classical physics failure: Classical theory predicted energy content depends on brightness, so both number and kinetic energy should depend on brightness. This contradicted observations (e.g., dim yellow light ejects electrons from potassium while bright red light doesn't, if red light's frequency is below threshold).

- Einstein (1905) explained photoelectric effect using Planck's quantum theory.

- o Light is viewed as a stream of particles called photons.

- o A photon with sufficient energy ($h\nu$) strikes an electron, transferring its energy instantaneously.

- o The minimum energy required to eject an electron is $h\nu_0$ (called work function, W_0).

- o The excess energy ($h\nu - h\nu_0$) is transferred as the kinetic energy (KE) of the photoelectron: $KE = \frac{1}{2} m_e v^2 = h\nu - h\nu_0$.

- o A more intense beam has more photons, thus ejecting more electrons.

D. Dual Behaviour of Electromagnetic Radiation

- The wave nature (explains interference, diffraction) and particle nature (explains black-body radiation, photoelectric effect) of light led to the conclusion that light possesses dual behaviour: it acts as both a wave and a stream of particles depending on the experiment.

E. Atomic Spectra (Evidence for Quantized Energy Levels)

- When white light passes through a prism, it disperses into a continuous spectrum (rainbow) because it contains all wavelengths.

- When atoms/molecules absorb energy and become excited,

they emit radiation to return to a lower energy state.

- The emission spectra of atoms in the gas phase are not continuous; they emit light only at specific wavelengths with dark spaces in between. These are called line spectra or atomic spectra.

- o Line emission spectra are unique to each element, acting like "fingerprints" for chemical analysis.

- Line Spectrum of Hydrogen:

- o When electric discharge passes through gaseous hydrogen, excited hydrogen atoms emit radiation of discrete frequencies.

- o The hydrogen spectrum consists of several series of lines.

- o Balmer (1885) found a formula for the visible lines (Balmer series) based on wavenumber ($\bar{\nu}$): $\bar{\nu} = 109,677 \text{ cm}^{-1} (1/2^2 - 1/n^2)$ where $n = 3, 4, 5, \dots$

- o Johannes Rydberg noted that all hydrogen spectral series could be described by: $\bar{\nu} = RH (1/n_1^2 - 1/n_2^2)$, where $n_2 > n_1$ and RH (Rydberg constant for hydrogen) = $109,677 \text{ cm}^{-1}$.

- o Different n_1 values correspond to different series:

- Lyman series: $n_1 = 1, n_2 = 2, 3, \dots$ (Ultraviolet region).

- Balmer series: $n_1 = 2, n_2 = 3, 4, \dots$ (Visible region).

- Paschen series: $n_1 = 3, n_2 = 4, 5, \dots$ (Infrared region).

- Brackett series: $n_1 = 4, n_2 = 5, 6, \dots$ (Infrared region).

- Pfund series: $n_1 = 5, n_2 = 6, 7, \dots$ (Infrared region).

- The line spectrum becomes more complex for heavier atoms, but the uniqueness and regularity of line spectra for each element suggested something about their electronic structure.

V. Bohr's Model for Hydrogen Atom (1913)

- Niels Bohr used Planck's concept of quantization to explain the hydrogen atom's structure and spectrum quantitatively.

- Postulates:

- o i) Fixed Orbits: Electrons move around the nucleus in specific circular paths of fixed radius and energy, called orbits,

stationary states, or allowed energy states. These orbits are concentric.

o ii) Quantized Energy Change: Electron energy in an orbit doesn't change over time. Energy is absorbed when an electron moves from a lower to a higher stationary state, and emitted when it moves from a higher to a lower state. This energy change is not continuous.

o iii) Bohr's Frequency Rule: The frequency (ν) of radiation absorbed or emitted during a transition between two stationary states (E_1 and E_2) is given by $\nu = (E_2 - E_1)/h = \Delta E/h$.

o iv) Quantized Angular Momentum: The angular momentum of an electron in a stationary state is quantized, expressed as $m_e v r = nh/2\pi$ (where $n = 1, 2, 3, \dots$). Only orbits where angular momentum is an integral multiple of $h/2\pi$ are allowed.

• Key Derivations for Hydrogen Atom:

o Stationary states are numbered by principal quantum numbers ($n = 1, 2, 3, \dots$).

o Radii of stationary states: $r_n = n^2 a_0$, where $a_0 = 52.9 \text{ pm}$ (Bohr orbit radius for $n=1$). Radius increases with increasing n .

o Energy of stationary states: $E_n = -RH/n^2$, where RH (Rydberg constant) $= 2.18 \times 10^{-18} \text{ J}$.

□ The negative sign of E_n means the electron's energy in the atom is lower than a free electron at rest (which has zero energy at infinite distance, $n=\infty$). More negative energy indicates greater stability ($n=1$ is the ground state, most stable).

• Application to Hydrogen-like Species: Bohr's theory can be applied to ions with only one electron (e.g., He^+ , Li^{2+} , Be^{3+}).

o Energy: $E_n = -2.18 \times 10^{-18} \text{ J} (Z^2/n^2)$.

o Radius: $r_n = 52.9 \text{ pm} (n^2/Z)$.

o Increased Z leads to more negative energy (tighter binding) and smaller radius.

• Explanation of Line Spectrum of Hydrogen:

- o Line spectra result from electrons transitioning between allowed energy orbits.
- o Absorption: Electron moves from smaller n to higher n (ΔE is positive).
- o Emission: Electron moves from higher n to lower n (ΔE is negative, energy released).
- o The energy gap (ΔE) between two orbits (n_i and n_f) is: $\Delta E = RH (1/n_i^2 - 1/n_f^2)$.
- o This expression is similar to Rydberg's empirical formula.

F. Limitations of Bohr's Model

- It was an improvement over Rutherford's model, explaining hydrogen stability and line spectra for hydrogen and hydrogen-like ions.
- However, it was too simple and failed to account for:
 - o Finer details (doublets) of the hydrogen spectrum seen with sophisticated techniques.
 - o Spectra of atoms other than hydrogen (e.g., helium with two electrons).
 - o Splitting of spectral lines in magnetic field (Zeeman effect) or electric field (Stark effect).
 - o The ability of atoms to form molecules by chemical bonds.
- Fundamental Flaws:
 - o Ignored the wave character of the electron.
 - o An orbit is a well-defined path, implying exact knowledge of both position and velocity simultaneously, which contradicts Heisenberg's Uncertainty Principle.

VI. Towards Quantum Mechanical Model of the Atom

- The shortcomings of Bohr's model led to new developments:
 - o Dual behaviour of matter.
 - o Heisenberg uncertainty principle.

A. Dual Behaviour of Matter

- Louis de Broglie (1924) proposed that matter, like radiation, also exhibits dual behaviour (both particle and wave-like properties).

- o Just as photons have momentum and wavelength, electrons should also have momentum and wavelength.

- o de Broglie relation: $\lambda = h/mv = h/p$ (where m is mass, v is velocity, p is momentum).

- o This prediction was confirmed by electron diffraction experiments, leading to tools like the electron microscope.

- The wave properties of ordinary macroscopic objects are not detectable due to their large masses, resulting in extremely short wavelengths.

- However, wavelengths associated with electrons and other subatomic particles (very small mass) can be detected.

B. Heisenberg's Uncertainty Principle

- Werner Heisenberg (1927) stated that it is impossible to simultaneously determine the exact position and exact momentum (or velocity) of an electron.

- o Mathematically: $\Delta x \Delta p_x \geq h/4\pi$ (where Δx is uncertainty in position, Δp_x is uncertainty in momentum).

- o If position (Δx) is known accurately (small), then velocity (Δv_x) will be highly uncertain (large), and vice versa.

- o This means physical measurements on electron position or velocity will always be "fuzzy" or "blur".

- Significance:

- o It rules out the existence of definite paths or trajectories for electrons and similar microscopic particles.

- o The classical concept of an electron moving in a well-defined orbit (like in Bohr's model) is not valid.

- o Its effect is significant only for microscopic objects like electrons and negligible for macroscopic objects.

- o Instead of precise position and momentum, quantum

mechanics talks about the probability of finding an electron at a given position and momentum.

VII. Quantum Mechanical Model of Atom

- Classical mechanics fails for microscopic objects because it ignores dual behavior and the uncertainty principle.
- Quantum mechanics is the branch of science that accounts for this dual behavior and specifies the laws of motion for microscopic objects.
- Developed independently by Werner Heisenberg and Erwin Schrödinger in 1926.

A. Schrödinger Equation

- The fundamental equation of quantum mechanics, developed by Schrödinger, incorporates de Broglie's wave-particle duality and is consistent with Heisenberg's uncertainty principle.
- For a system where energy doesn't change with time (like an atom), the Schrödinger equation's solution gives the possible energy levels an electron can occupy and the corresponding wave function (ψ) of the electron.
- These quantized energy states and wave functions are characterized by a set of three quantum numbers: principal quantum number (n), azimuthal quantum number (l), and magnetic quantum number (m_l).
- The wave function (ψ) itself has no physical meaning.
- Atomic orbitals: Wave functions of hydrogen or hydrogen-like species (one-electron systems).
- The probability of finding an electron at a point within an atom is proportional to $|\psi|^2$ at that point. $|\psi|^2$ is known as probability density.

B. Important Features of the Quantum Mechanical Model

- 1. Quantized Electron Energy: Electron energies in atoms are quantized, meaning they can only have specific discrete values when bound to the nucleus.

- 2. Origin of Quantization: Quantized electronic energy levels are a direct result of the wave-like properties of electrons and are allowed solutions of the Schrödinger wave equation.
- 3. Heisenberg Uncertainty Principle Applied: The exact position and velocity of an electron cannot be simultaneously determined, so its path in an atom cannot be accurately known. Instead, we speak of the probability of finding the electron at different points.
- 4. Atomic Orbitals: An atomic orbital is the wave function (ψ) for an electron in an atom.
 - o Many atomic orbitals are possible, each with a definite energy.
 - o An orbital cannot contain more than two electrons.
 - o In multi-electron atoms, electrons fill orbitals in order of increasing energy.
 - o The orbital wave function (ψ) contains all information about the electron in that orbital.
- 5. Probability Density: The probability of finding an electron at a point is proportional to the square of the orbital wave function, $|\psi|^2$. This is called probability density and is always positive. It helps predict the region where the electron will most probably be found.

C. Orbitals and Quantum Numbers

- Orbitals are distinguished by their size, shape, and orientation.
- Each orbital is designated by three quantum numbers: n , l , and m_l . A fourth quantum number, electron spin (m_s), was later proposed.

1. Principal Quantum Number (n)

- A positive integer: $n = 1, 2, 3, \dots$
- Determines the size and, to a large extent, the energy of the orbital.

- For hydrogen and hydrogen-like species, energy and size depend only on n .
- Identifies the shell (K, L, M, N for $n=1, 2, 3, 4$ respectively).
- The number of allowed orbitals for a given n is n^2 .
- Orbital size and energy increase with increasing n (electron is further from the nucleus).

2. Azimuthal (or Orbital Angular Momentum or Subsidiary) Quantum Number (l)

- Defines the three-dimensional shape of the orbital.
- For a given n , l can have n values from 0 to $n-1$.
 - o e.g., if $n=1$, $l=0$. If $n=2$, $l=0, 1$. If $n=3$, $l=0, 1, 2$.
- Each shell consists of n sub-shells (or sub-levels).
- Notation for sub-shells based on l values:
 - o $l = 0$: s orbital
 - o $l = 1$: p orbital
 - o $l = 2$: d orbital
 - o $l = 3$: f orbital
 - o $l = 4$: g orbital
 - o $l = 5$: h orbital

3. Magnetic Orbital Quantum Number (m_l)

- Gives information about the spatial orientation of the orbital.
- For any sub-shell (given l value), there are $(2l+1)$ possible m_l values: $-l, -(l-1), \dots, 0, \dots, (l-1), l$.
 - o $l=0$ (s): $m_l = 0$ (1 s orbital).
 - o $l=1$ (p): $m_l = -1, 0, +1$ (3 p orbitals).
 - o $l=2$ (d): $m_l = -2, -1, 0, +1, +2$ (5 d orbitals).
- Thus, s subshell has 1 orbital, p has 3, d has 5, f has 7, etc..
- m_l values relate to orientation (e.g., $2p_x, 2p_y, 2p_z$ lobes along x, y, z axes).

4. Electron Spin Quantum Number (m_s)

- Proposed by George Uhlenbeck and Samuel Goudsmit in 1925.
- Describes the intrinsic spin angular momentum of an electron

around its own axis.

- Can have two orientations relative to a chosen axis, represented by $m_s = +\frac{1}{2}$ or $-\frac{1}{2}$.
- These are called spin up ($\uparrow$) and spin down ($\downarrow$).
- An orbital can hold a maximum of two electrons, and they must have opposite spins.

5. Summary of Quantum Number Information:

- n : Defines the shell, determines orbital size, and largely energy.
- l : Identifies the subshell, determines orbital shape, and to some extent, energy in multi-electron atoms. There are n subshells in the n th shell, and $(2l+1)$ orbitals of each type in a subshell.
- m_l : Designates the orientation of the orbital. For a given l , there are $(2l+1)$ values, equal to the number of orbitals per subshell.
- m_s : Refers to the orientation of the electron's spin.

6. Orbit vs. Orbital:

- Orbit (Bohr): A circular path around the nucleus where an electron moves. It's a precise, defined path. Bohr orbits have no real meaning as precise paths are impossible due to the Heisenberg Uncertainty Principle.
- Orbital (Quantum Mechanical): A one-electron wave function (ψ) in an atom. It is characterized by n , l , and m_l . ψ has no physical meaning itself, but $|\psi|^2$ (probability density) at any point gives the probability of finding the electron at that point.
 - o Probability density is probability per unit volume.
 - o Boundary surface diagrams enclose a region (e.g., 90% probability) where the electron is most likely found.
- Shapes of Atomic Orbitals:
 - o s -orbitals ($l=0$): Spherically symmetric. Size increases with n .

($4s > 3s > 2s > 1s$). Probability density is maximum at the nucleus for $1s$, but $2s$ and higher s -orbitals have nodal surfaces/nodes (regions where probability density is zero). ns -orbital has $(n-1)$ nodes.

o p -orbitals ($l=1$): Appear as two lobes on either side of a plane passing through the nucleus (dumbbell shape). There are three p -orbitals (p_x, p_y, p_z), mutually perpendicular. Size and energy increase with n ($4p > 3p > 2p$). They have $(n-2)$ radial nodes and one angular node (the plane through the nucleus).

o d -orbitals ($l=2$): Minimum n value is 3. There are five d -orbitals ($d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$). The first four are similar, d_{z^2} is different, but all five are equivalent in energy for a given n . They have $(n-3)$ radial nodes and two angular nodes.

o Total number of nodes = $(n-1)$, which is the sum of l angular nodes and $(n-l-1)$ radial nodes.

D. Energies of Orbitals

• Hydrogen Atom: Electron energy depends solely on the principal quantum number (n). Orbitals with the same n value have the same energy; they are degenerate (e.g., $2s = 2p$).

• Multi-electron Atoms: Electron energy depends on both the principal quantum number (n) and the azimuthal quantum number (l).

o For a given n , energy increases in the order $s < p < d < f$.

o This difference arises from mutual repulsion among electrons and shielding effect.

o Shielding: Inner shell electrons partially "screen" the outer shell electrons from the full nuclear positive charge (Ze), reducing the effective nuclear charge (Z_{eff}) experienced by outer electrons.

o Penetration: s -orbitals penetrate closer to the nucleus and shield more effectively than p -orbitals, which shield more than

d-orbitals. Thus, for a given shell, Z_{eff} experienced by the electron decreases with increasing l .

o This leads to the splitting of energy levels within the same shell.

o **$(n+l)$ rule**: The lower the $(n+l)$ value for an orbital, the lower its energy. If $(n+l)$ values are the same, the orbital with the lower n value has lower energy.

o Examples of staggering: $4s < 3d$ and $6s < 5d$.

o Energies of orbitals in the same subshell decrease with increasing atomic number (Z_{eff}) (e.g., $E_{2s}(\text{H}) > E_{2s}(\text{Li}) > E_{2s}(\text{Na})$).

E. Filling of Orbitals in Atoms (Electronic Configuration)

• Electrons fill orbitals according to three rules:

o 1. Aufbau Principle: "Building up" principle. In the ground state, orbitals are filled in order of increasing energies.

Electrons occupy the lowest energy orbitals first.

□ Useful energy order: $1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 4f, 5d, 6p, 7s, \dots$. This is a general guide, and exceptions can occur.

o 2. Pauli Exclusion Principle: No two electrons in an atom can have the same set of four quantum numbers.

□ Alternatively: "Only two electrons may exist in the same orbital, and these electrons must have opposite spin".

□ This limits the capacity of subshells: $1s$ (2 electrons), p (6 electrons), d (10 electrons), f (14 electrons).

□ Maximum electrons in a shell with principal quantum number n is $2n^2$.

o 3. Hund's Rule of Maximum Multiplicity: For orbitals belonging to the same subshell (degenerate orbitals), pairing of electrons does not occur until each orbital in that subshell has one electron, and these single electrons have parallel spins.

□ Pairing starts with the 4th electron in p orbitals, 6th in d ,

and 8th in f.

□ Half-filled and fully-filled degenerate sets of orbitals acquire extra stability due to symmetry.

F. Electronic Configuration of Atoms

• Electronic Configuration: The distribution of electrons into orbitals of an atom.

• Notation:

o i) $s^a p^b d^c \dots$ notation: Subshell symbol with superscript for number of electrons. Principal quantum number precedes the subshell (e.g., $1s^2$, $2p^6$).

o ii) Orbital diagram: Each orbital represented by a box, electrons by arrows ($\uparrow$ for $+\frac{1}{2}$ spin, $\downarrow$ for $-\frac{1}{2}$ spin). This notation represents all four quantum numbers.

• Examples:

o H: $1s^1$

o He: $1s^2$ (two electrons with opposite spins in $1s$ orbital).

o Li: $1s^2 2s^1$ (3rd electron in $2s$ due to Pauli exclusion).

o B ($1s^2 2s^2 2p^1$), C ($1s^2 2s^2 2p^2$), N ($1s^2 2s^2 2p^3$), O ($1s^2 2s^2 2p^4$), F ($1s^2 2s^2 2p^5$), Ne ($1s^2 2s^2 2p^6$) - $2p$ orbitals fill progressively.

o Na to Ar: $[\text{Ne}]3s^1$ to $[\text{Ne}]3s^2 3p^6$ (filling $3s$ and $3p$ orbitals).

o Core electrons: Electrons in completely filled inner shells (e.g., Ne in $[\text{Ne}]$).

o Valence electrons: Electrons in the outermost shell (highest principal quantum number).

o K ($[\text{Ar}]4s^1$) and Ca ($[\text{Ar}]4s^2$): $4s$ orbital fills before $3d$ ($4s$ is lower in energy than $3d$).

o Sc to Zn: $3d$ orbitals fill progressively (e.g., Sc ($[\text{Ar}]3d^1 4s^2$)).

o Exceptions (Cr, Cu): Chromium and Copper show d^5 and d^{10} configurations respectively, instead of d^4 and d^9 , because fully-filled and half-filled orbitals have extra stability. (e.g., Cr: $[\text{Ar}]3d^5 4s^1$, Cu: $[\text{Ar}]3d^{10} 4s^1$).

- Understanding electronic configuration is essential for explaining chemical behavior (e.g., why atoms combine, metallic/non-metallic properties, reactivity).

G. Stability of Completely Filled and Half-filled Subshells

- Ground state configuration corresponds to lowest total electronic energy.

- Extra stability for half-filled (p^3 , d^5 , f^7) and completely filled (p^6 , d^{10} , f^{14}) subshells.

- Reasons for Stability:

- o 1. Symmetrical distribution of electrons: Symmetry leads to stability. These configurations have symmetrical electron distribution. Electrons in the same subshell have equal energy but different spatial distributions, leading to relatively small shielding and stronger attraction by the nucleus.

- o 2. Exchange Energy: A stabilizing effect when two or more electrons with the same spin are in degenerate orbitals. These electrons tend to exchange positions, and the energy released from this exchange is called exchange energy. The number of possible exchanges is maximum when the subshell is half-filled or completely filled, maximizing exchange energy and stability. This phenomenon is the basis of Hund's rule, favoring parallel spins in degenerate orbitals.